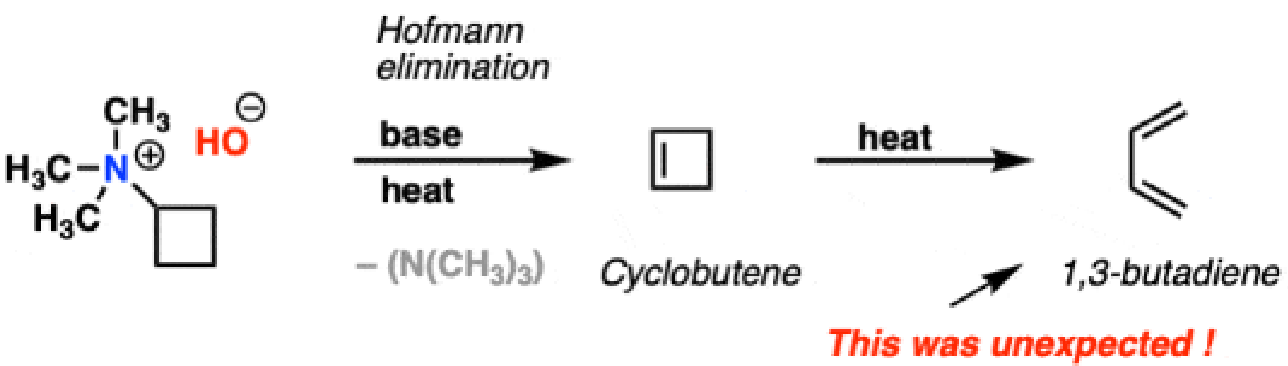
History of Pericyclic Reactions

The year was 1905. Chemist Richard Willstätter and his collaborator Wolfgang von Schmaedel were heating a reaction mixture in their laboratory at the University of Munich. They were performing a fairly routine elimination reaction that has now become known as the Hofmann elimination.

The expected product was cyclobutene.

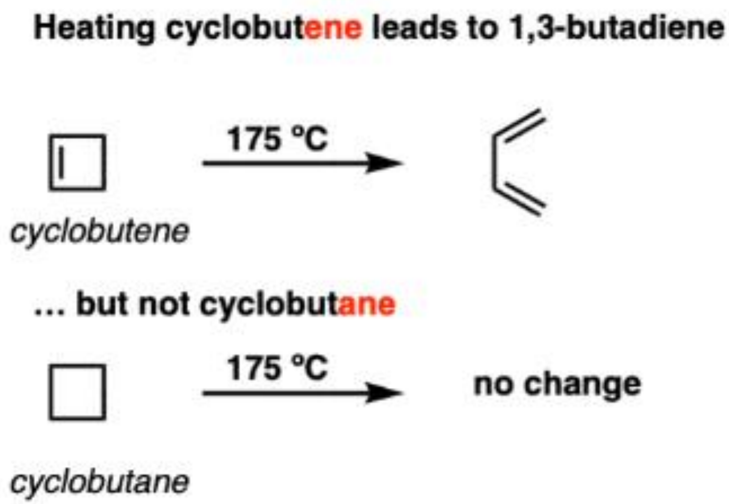
When the reaction mixture was analyzed, cyclobutene was indeed formed. Mysteriously, however, considerable amounts of 1,3-butadiene were also isolated.

Willstatter's accidental discovery of electrocyclic ring-opening (1905)



History of Pericyclic Reactions

In the following decades, the reaction (and its many variants) was studied in more detail. It was verified that heating cyclobutene did indeed form 1,3-butadiene. Interestingly, however, heating cyclobutane did not lead to any ring-opening reaction.



Pericyclic Reactions

Although most organic reactions take place by way of ionic or radical intermediates, a number of useful reactions occur in one-step processes that do *not* form reactive intermediates.

- A pericyclic reaction is a concerted reaction that proceeds through a cyclic transition state.

Pericyclic reactions require light or heat and are completely stereospecific; that is, a single stereoisomer of the reactant forms a single stereoisomer of the product. We will consider two categories of pericyclic reactions: **electrocyclic reactions** and **cycloadditions.**

An electrocyclic reaction is a reversible reaction that can involve ring closure or ring opening.

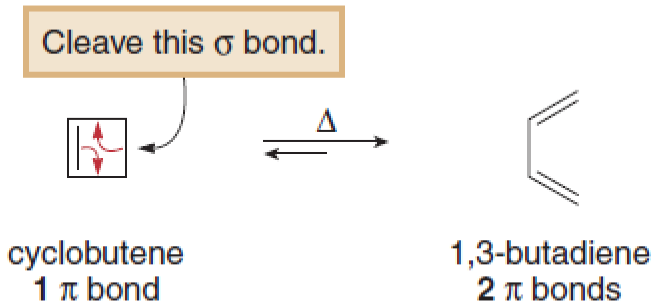
- An electrocyclic ring closure is an intramolecular reaction that forms a cyclic product containing one more σ bond and one fewer π bond than the reactant.

1,3,5-hexatriene
3 π bonds

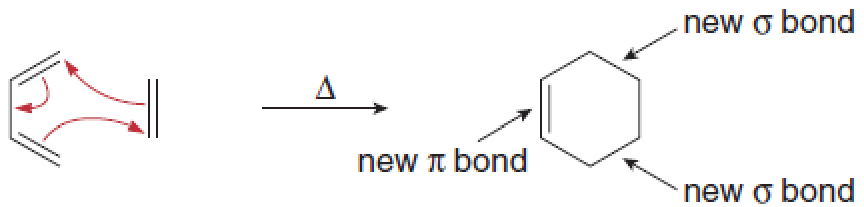
1,3-cyclohexadiene
2 π bonds

Electrocyclic Reactions

- An electrocyclic ring-opening reaction is a reaction in which a σ bond of a cyclic reactant is cleaved to form a conjugated product with one more π bond.



- A cycloaddition is a reaction between two compounds with π bonds to form a cyclic product with two new σ bonds.



Electrocyclic Reactions

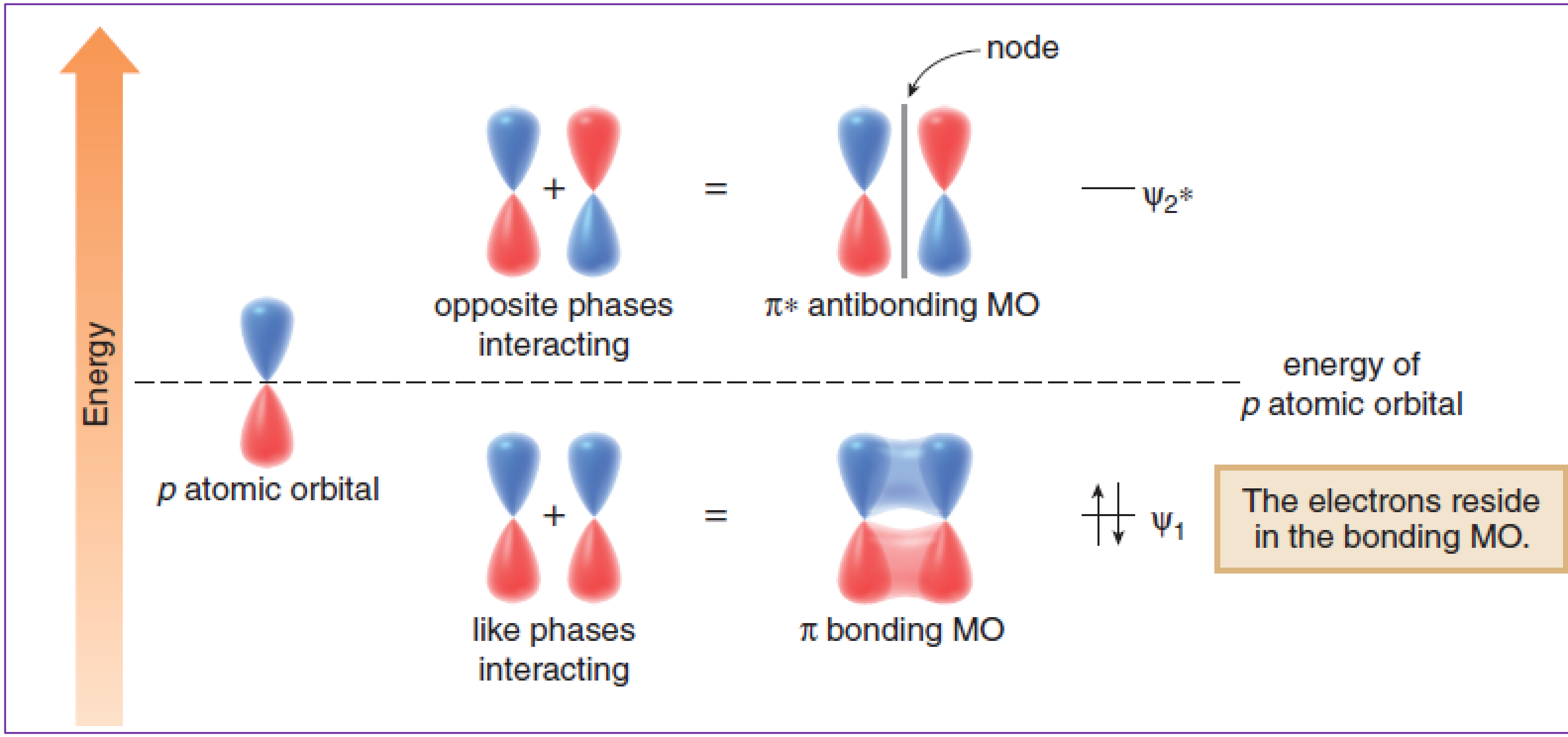
Two features determine the course of the reactions: the number of π bonds involved and whether the reaction occurs in the presence of heat (thermal conditions) or light (photochemical conditions). These reactions follow a set of rules based on orbitals and symmetry first proposed by R. B. Woodward and Roald Hoffmann in 1965, and derived from theory described by Kenichi Fukui in 1954.

Background on Molecular Orbitals

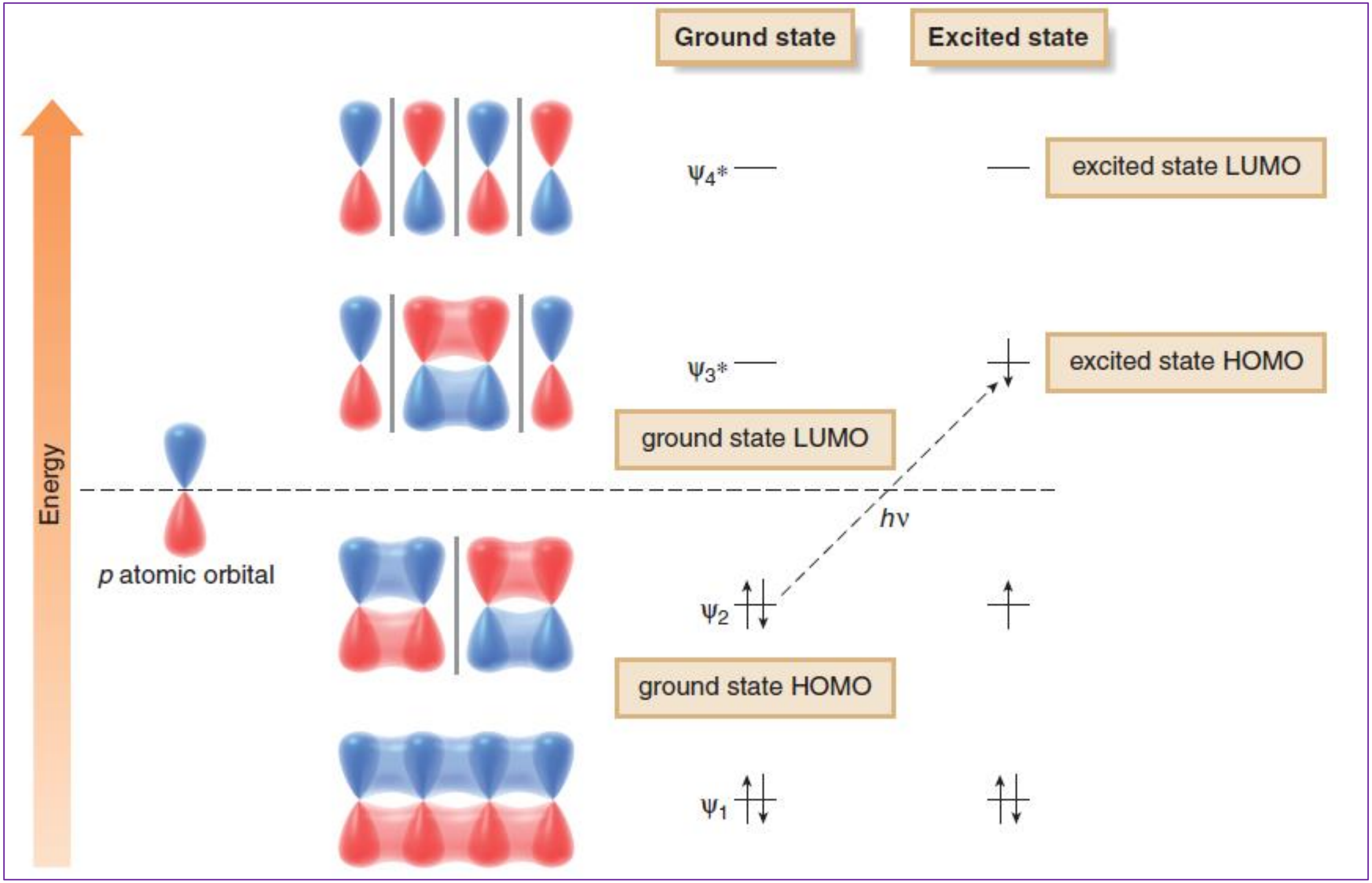
Molecular orbital (MO) theory describes bonds as the mathematical combination of atomic orbitals that forms a new set of orbitals called **molecular orbitals (MOs). The number of atomic orbitals used *equals* the number of molecular orbitals formed.**

Since pericyclic reactions involve π bonds, let's examine the molecular orbitals that result from *p* orbital overlap in ethylene, 1,3-butadiene, and 1,3,5-hexatriene, molecules that contain one, two, and three π bonds, respectively. Keep in mind that the two lobes of a *p* orbital are opposite in phase, with a node of electron density at the nucleus.

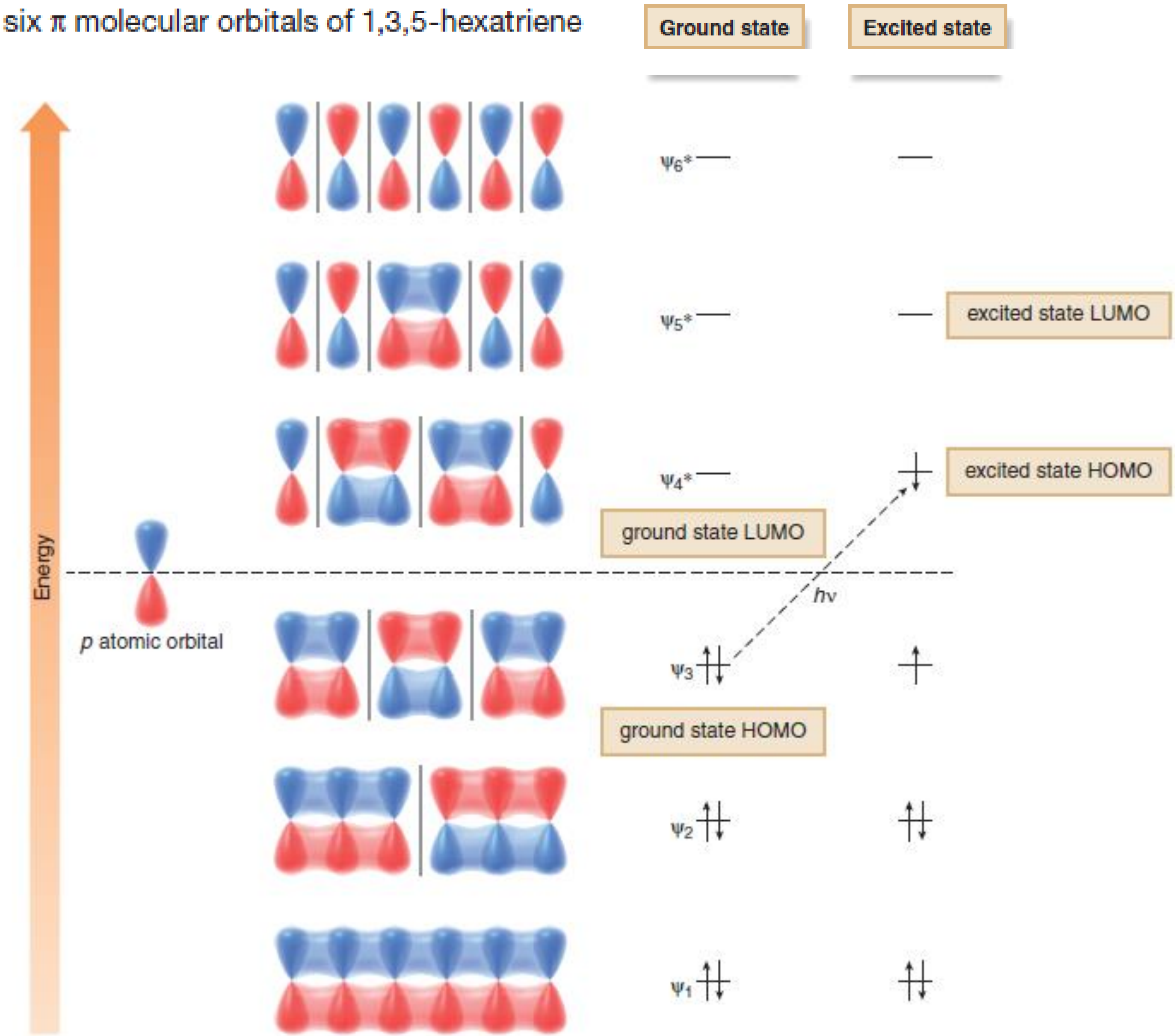
The π and π* molecular orbitals of ethylene



The four π molecular orbitals of 1,3-butadiene



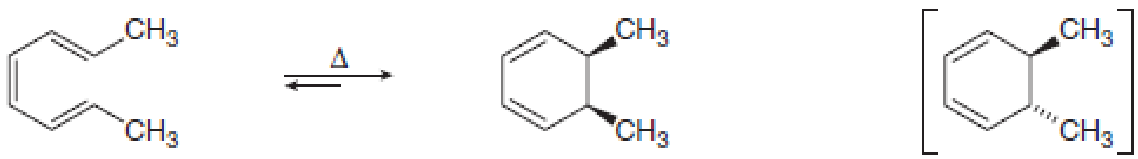
The six π molecular orbitals of 1,3,5-hexatriene



Electrocyclic Reactions

Stereochemistry and Orbital Symmetry

Electrocyclic reactions are completely stereospecific. For example, ring closure of (2*E*,4*Z*,6*E*)-2,4,6-octatriene yields a single product with cis methyl groups on the ring. Ring opening of *cis*-3,4-dimethylcyclobutene forms a single conjugated diene with one *Z* alkene and one *E* alkene.

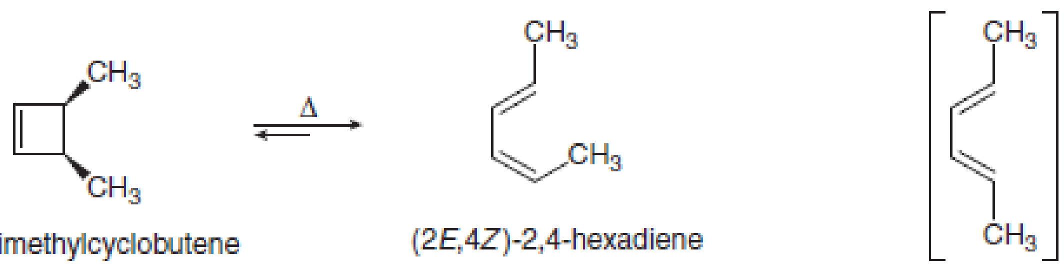


(2*E*,4*Z*,6*E*)-2,4,6-octatriene

cis-5,6-dimethyl-1,3-cyclohexadiene

NOT formed

cis product only



cis-3,4-dimethylcyclobutene

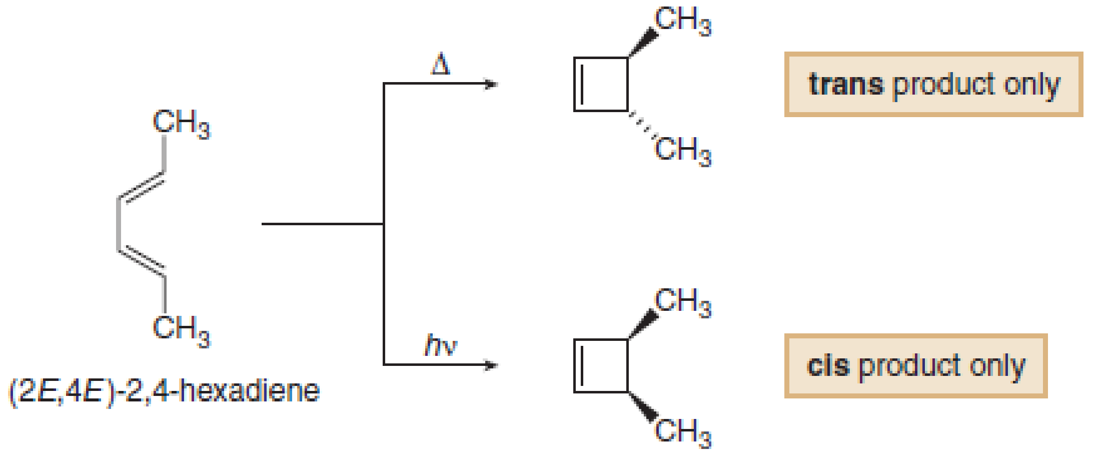
(2*E*,4*Z*)-2,4-hexadiene

NOT formed

(2*E*,4*Z*) diene only

Electrocyclic Reactions

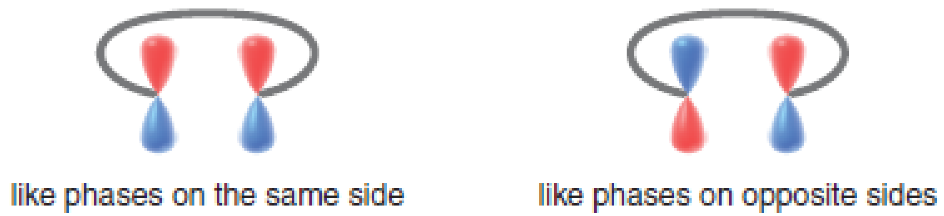
Moreover, the stereochemistry of the product of an electrocyclic reaction depends on whether the reaction is carried out under thermal or photochemical reaction conditions—that is, with heat or light, respectively. Cyclization of (2*E*,4*E*)-2,4-hexadiene with heat forms a cyclobutene with trans methyl groups, whereas cyclization with light forms a cyclobutene with cis methyl groups.



Electrocyclic ring closure generally forms either an achiral meso compound or a mixture of chiral enantiomers. When enantiomers form, only one enantiomer is drawn in these reactions.

Electrocyclic Reactions

To understand these results, we must focus on the HOMO of the acyclic conjugated polyene that is either the reactant or product in an electrocyclic reaction. In particular, we must examine the *p* orbitals on the terminal carbons of the HOMO, and determine whether like phases of the orbitals are on the same side or on opposite sides of the molecule.

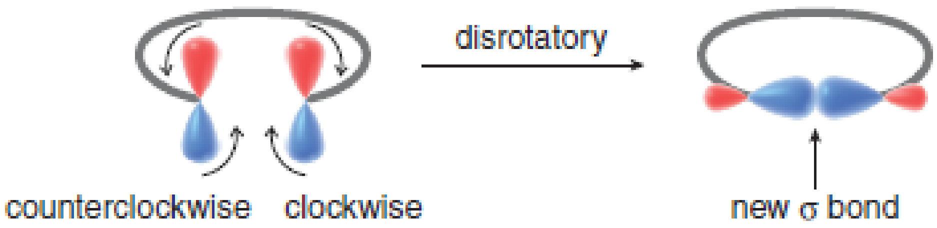


- An electrocyclic reaction occurs only when like phases of orbitals can overlap to form a bond. Such a reaction is *symmetry allowed.*
- An electrocyclic reaction cannot occur between orbitals of opposite phase. Such a reaction is *symmetry forbidden.*

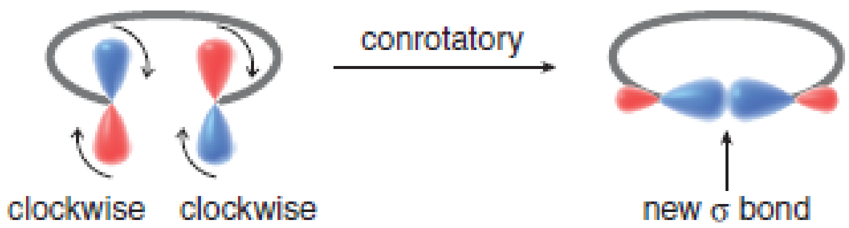
To form a bond, the *p* orbitals on the terminal carbons must rotate so that like phases can interact to form the new σ bond. Two modes of rotation are possible.

Electrocyclic Reactions

- When like phases of the *p* orbitals are on the same side of the molecule, the two orbitals must rotate in opposite directions—one clockwise and one counterclockwise. Rotation in opposite directions is said to be *disrotatory*.



- When like phases of the *p* orbitals are on opposite sides of the molecule, the two orbitals must rotate in the same direction—both clockwise or both counterclockwise. Rotation in the same direction is said to be *conrotatory*.



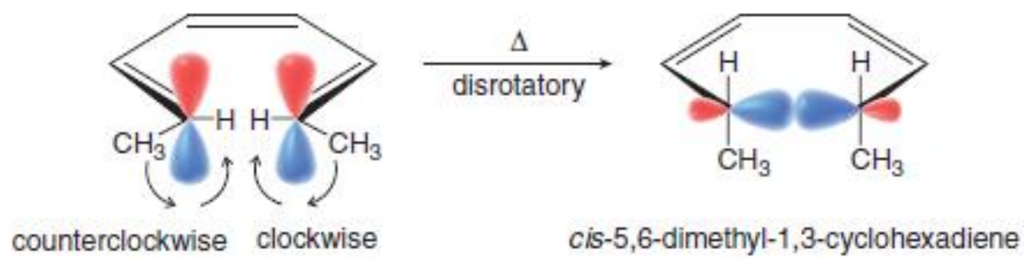
Thermal Electrocyclic Reactions

To explain the stereochemistry observed in electrocyclic reactions, we must examine the symmetry of the molecular orbital that contains the most loosely held π electrons. In a thermal reaction, we consider the **HOMO of the ground state electronic configuration.** Rotation occurs in a disrotatory or conrotatory fashion so that like phases of the *p* orbitals on the terminal carbons of this molecular orbital combine.

- **The number of double bonds in the conjugated polyene determines whether rotation is conrotatory or disrotatory.**

Two examples illustrate different outcomes.

Thermal electrocyclic ring closure of (2*E*,4*Z*,6*E*)-2,4,6-octatriene yields a single product with cis methyl groups on the ring.



(2*E*,4*Z*,6*E*)-2,4,6-octatriene

cis-5,6-dimethyl-1,3-cyclohexadiene

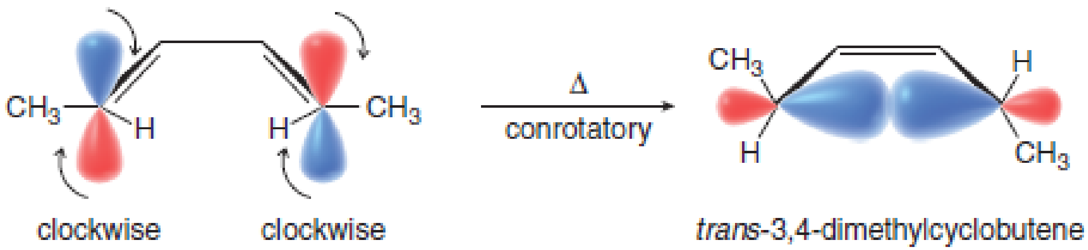
cis product

This is a specific example of the general process observed for conjugated polyenes with an *odd* number of π bonds. The HOMO of a conjugated polyene with an odd number of π bonds has like phases of the outermost *p* orbitals on the *same* side of the molecule. As a result:

- **Thermal electrocyclic reactions occur in a *disrotatory* fashion for a conjugated polyene with an *odd* number of π bonds.**

Electrocyclic Reactions

In contrast, thermal electrocyclic ring closure of (2*E*,4*E*)-2,4-hexadiene forms a cyclobutene with trans methyl groups.



(2*E*,4*E*)-2,4-hexadiene

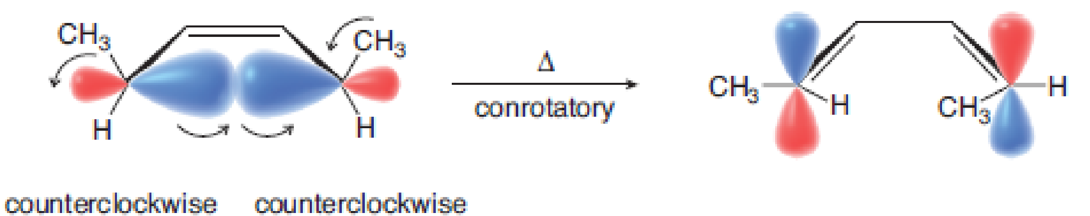
trans-3,4-dimethylcyclobutene
trans product

This is a specific example of the general process observed for conjugated polyenes with an *even* number of π bonds. The HOMO of a conjugated polyene with an even number of π bonds has like phases of the outermost *p* orbitals on *opposite* sides of the molecule. As a result:

- **Thermal electrocyclic reactions occur in a *conrotatory* fashion for a conjugated polyene with an *even* number of π bonds.**

Since electrocyclic reactions are reversible, **electrocyclic ring-opening reactions follow the same rules** as electrocyclic ring closures. Thus, thermal ring opening of *cis*-3,4-dimethylcyclobutene—

which ring opens to a diene with an *even* number of π bonds—occurs in a conrotatory fashion to form (2*E*,4*Z*)-2,4-hexadiene as the only product.



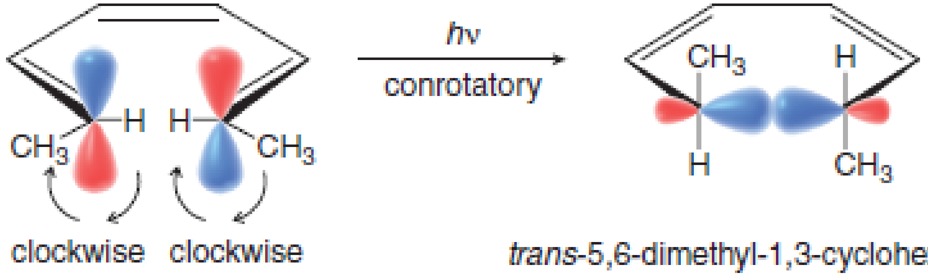
cis-3,4-dimethylcyclobutene

(2*E*,4*Z*)-2,4-hexadiene

Photochemical Electrocyclic Reactions

Photochemical electrocyclic reactions follow similar principles as those detailed in thermal reactions with one important difference. **In photochemical reactions, we must consider the orbitals of the HOMO of the excited state to determine the course of the reaction.** The excited state HOMO has the *opposite* orientation of the outermost *p* orbitals compared to the HOMO of the ground state. As a result, **the method of ring closure of a photochemical electrocyclic reaction is *opposite* to that of a thermal electrocyclic reaction for the same number of π bonds.**

Photochemical electrocyclic ring closure of (2*E*,4*Z*,6*E*)-2,4,6-octatriene yields a cyclic product with trans methyl groups on the ring.



(2*E*,4*Z*,6*E*)-2,4,6-octatriene

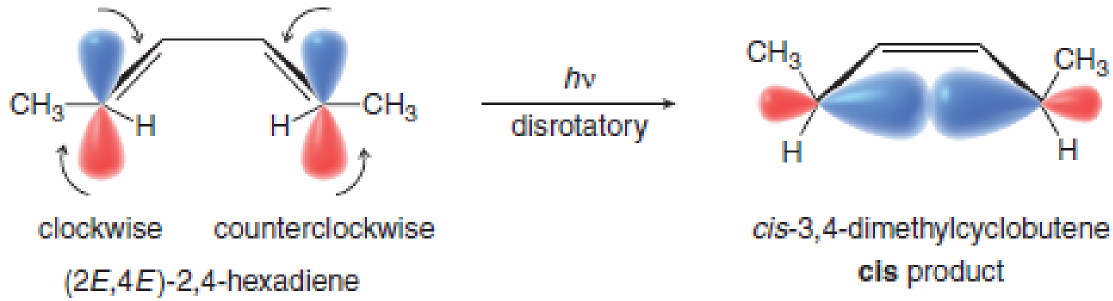
trans-5,6-dimethyl-1,3-cyclohexadiene

trans product

- Photochemical electrocyclic reactions occur in a *conrotatory* fashion for a conjugated polyene with an *odd* number of π bonds.

Photochemical Electrocyclic Reactions

Photochemical electrocyclic ring closure of (2*E*,4*E*)-2,4-hexadiene forms a cyclobutene with cis methyl groups.

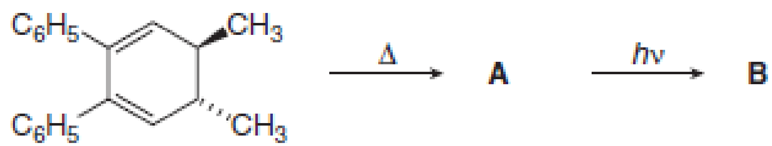


- Photochemical electrocyclic reactions occur in a *disrotatory* fashion for a conjugated polyene with an *even* number of π bonds.

Woodward–Hoffmann rules for electrocyclic reactions

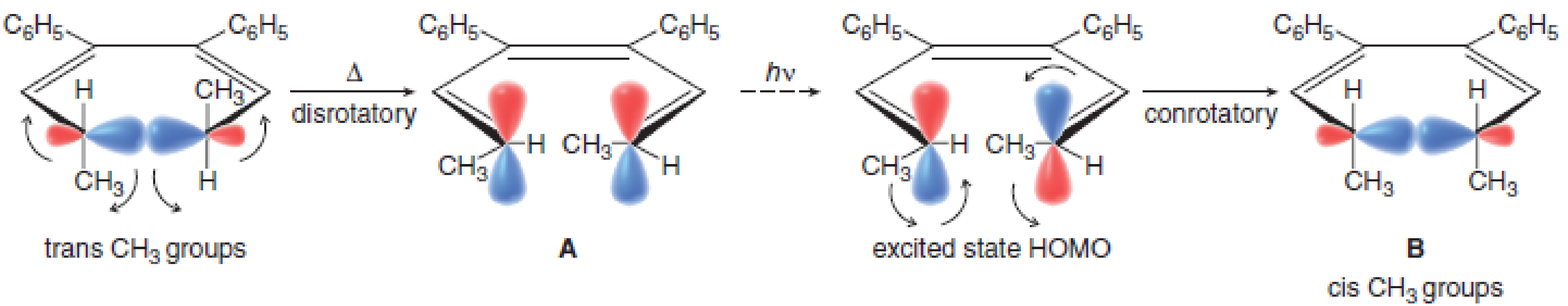
Number of π bonds	Thermal reaction	Photochemical reaction
Even	Conrotatory	Disrotatory
Odd	Disrotatory	Conrotatory

Identify **A** and **B** in the following reaction sequence. Label each process as conrotatory or disrotatory.



Solution

Ring opening of a cyclohexadiene forms a hexatriene with **three** π bonds. A conjugated polyene with an odd number of π bonds undergoes a thermal electrocyclic reaction in a disrotatory fashion (Table C.1). The resulting hexatriene (**A**) then undergoes a photochemical electrocyclic reaction in a conrotatory fashion to form a cyclohexadiene with cis methyl groups (**B**).



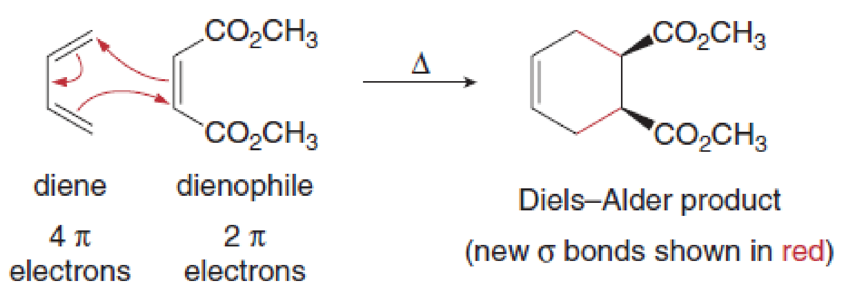
Cycloaddition Reactions

A cycloaddition is a reaction between two compounds with π bonds to form a cyclic product with two new σ bonds. Like electrocyclic reactions, cycloadditions are concerted, stereospecific reactions, and the course of the reaction is determined by the symmetry of the molecular orbitals of the reactants.

Cycloadditions can be initiated by heat (thermal conditions) or light (photochemical conditions). Cycloadditions are identified by the number of π electrons in the two reactants.

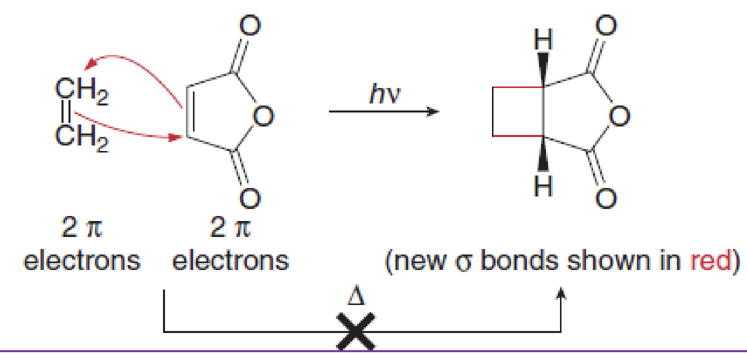
The Diels–Alder reaction is a thermal [4 + 2] cycloaddition that occurs between a diene with four π electrons and an alkene (dienophile) with two π electrons (Sections 16.12–16.14).

[4 + 2] Cycloaddition



A photochemical [2 + 2] cycloaddition occurs between two alkenes, each with two π electrons, to form a cyclobutane. Thermal [2 + 2] cycloadditions do *not* take place.

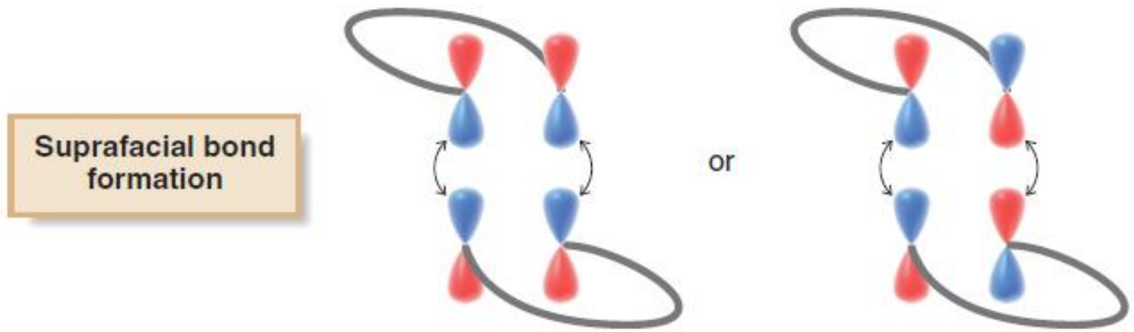
[2 + 2] Cycloaddition



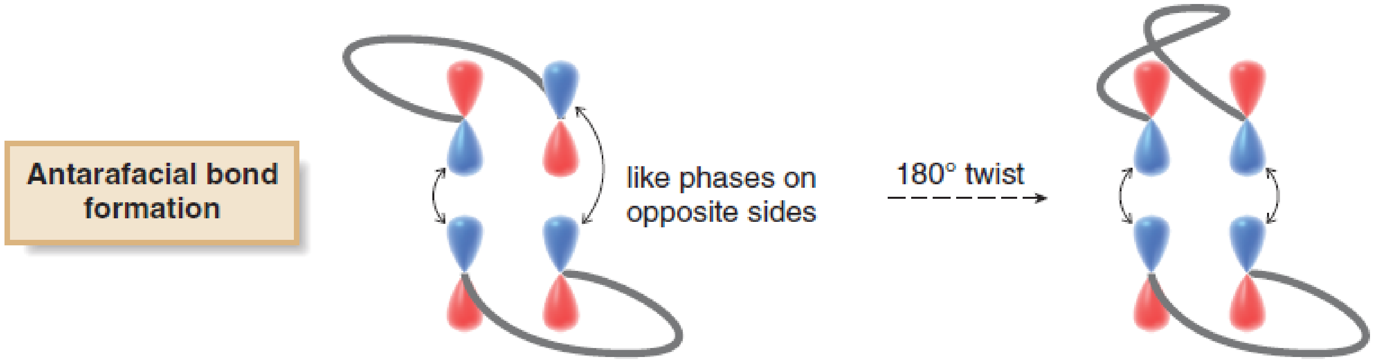
Orbital Symmetry and Cycloadditions

To understand cycloaddition reactions, we examine the *p* orbitals of the terminal carbons of both reactants. Bonding can take place only when like phases of both sets of *p* orbitals can combine. Two modes of reaction are possible.

- A suprafacial cycloaddition occurs when like phases of the *p* orbitals of both reactants are on the same side of the π system, so that two bonding interactions result.



- An antarafacial cycloaddition occurs when one π system must twist to align like phases of the *p* orbitals of the terminal carbons of the reactants.



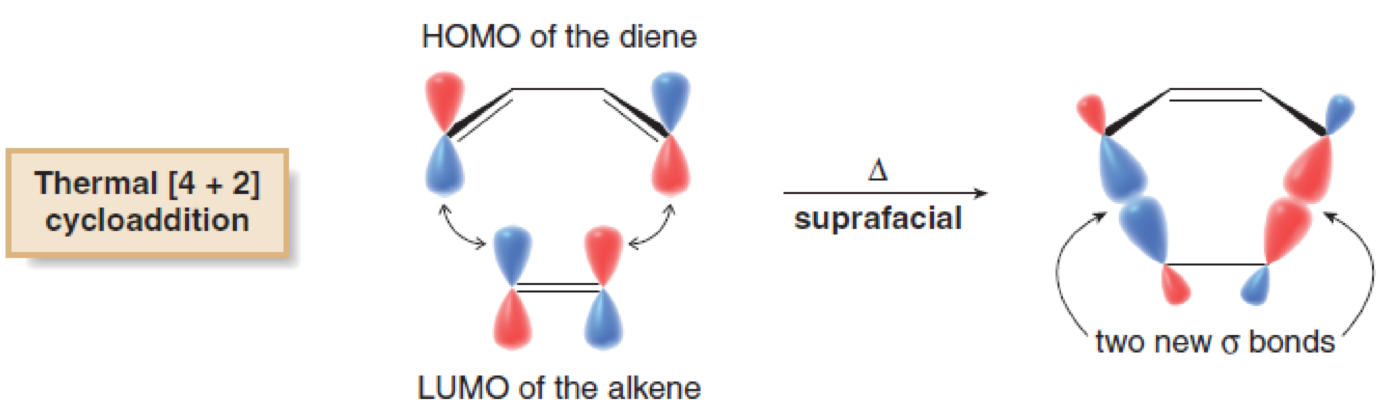
Because of the geometrical constraints of small rings, **cycloadditions that form four- or six-membered rings must take place by suprafacial pathways.**

Since cycloaddition involves the donation of electron density from one reactant to another, one reactant donates its most loosely held electrons—those occupying its HOMO—to a vacant orbital that can accept electrons—the LUMO—of the second reactant. The HOMO of either reactant can be used for analysis.

- In a cycloaddition we examine the bonding interactions of the HOMO of one component with the LUMO of the second component.

[4 + 2] Cycloadditions

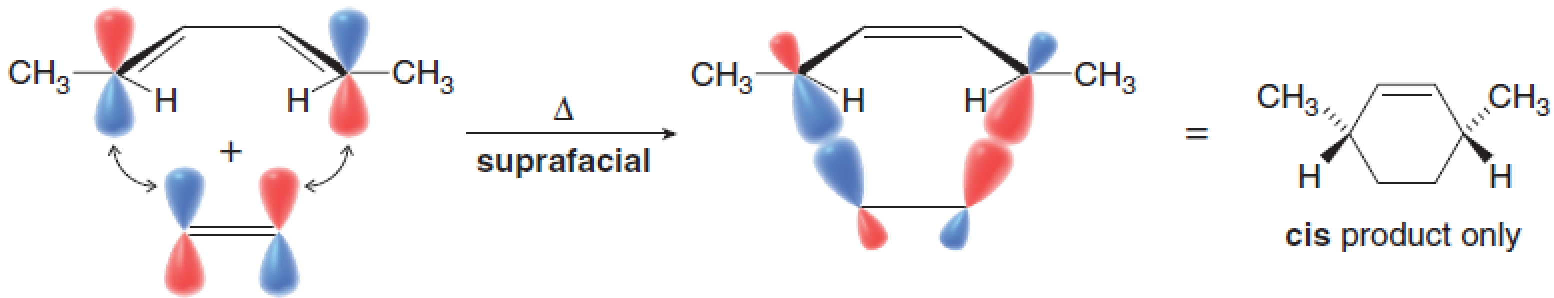
To examine the course of a [4 + 2] cycloaddition, let's arbitrarily choose the HOMO of the diene and the LUMO of the alkene, and look at the symmetry of the *p* orbitals on the terminal carbons of both components. Since two bonding interactions result from overlap of the like phases of both sets of *p* orbitals, **a [4 + 2] cycloaddition occurs readily by suprafacial reaction under thermal conditions.**



- Thermal cycloadditions involving an *odd* number of π bonds proceed by a *suprafacial* pathway.

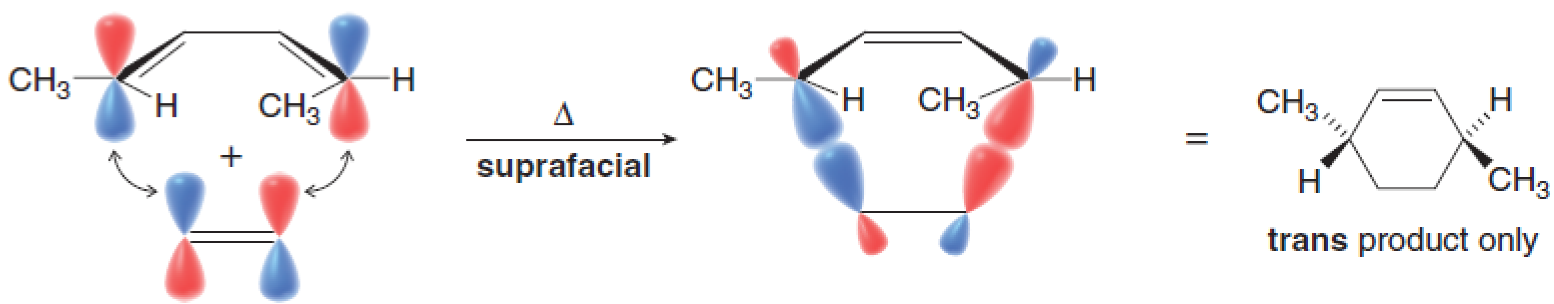
Because a Diels–Alder reaction follows a concerted, suprafacial pathway, the **stereochemistry of the diene is retained in the Diels–Alder product.** As a result, reaction of (2*E*,4*E*)-2,4-hexadiene with ethylene forms a cyclohexene with cis substituents (Reaction [1]), whereas reaction of (2*E*,4*Z*)-2,4-hexadiene with ethylene forms a cyclohexene with trans substituents (Reaction [2]).

[1]



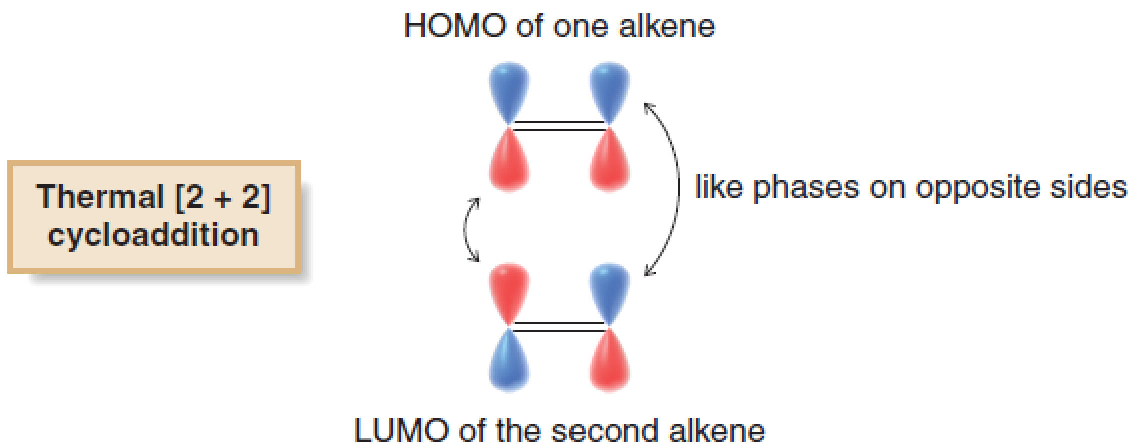
(2*E*,4*E*)-2,4-hexadiene

[2]

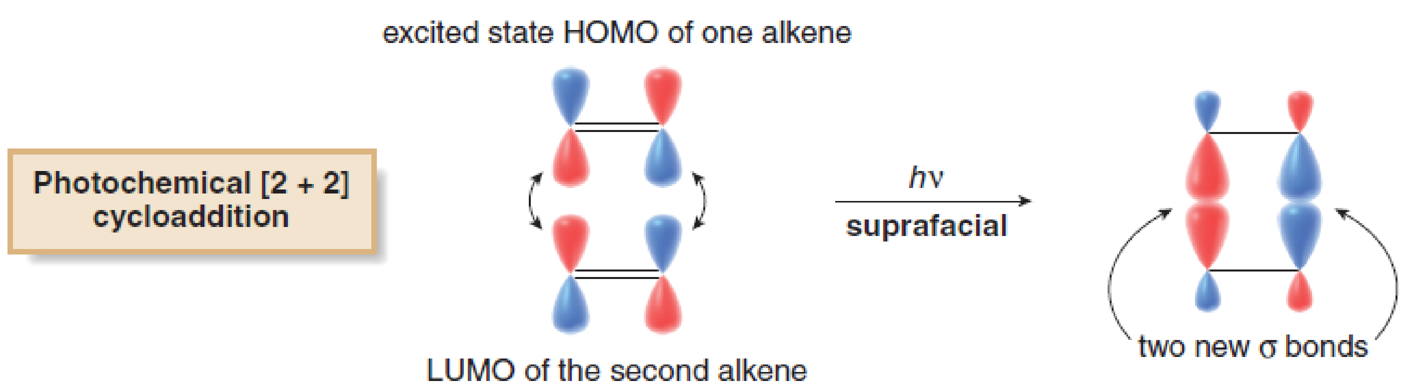


[2 + 2] Cycloadditions

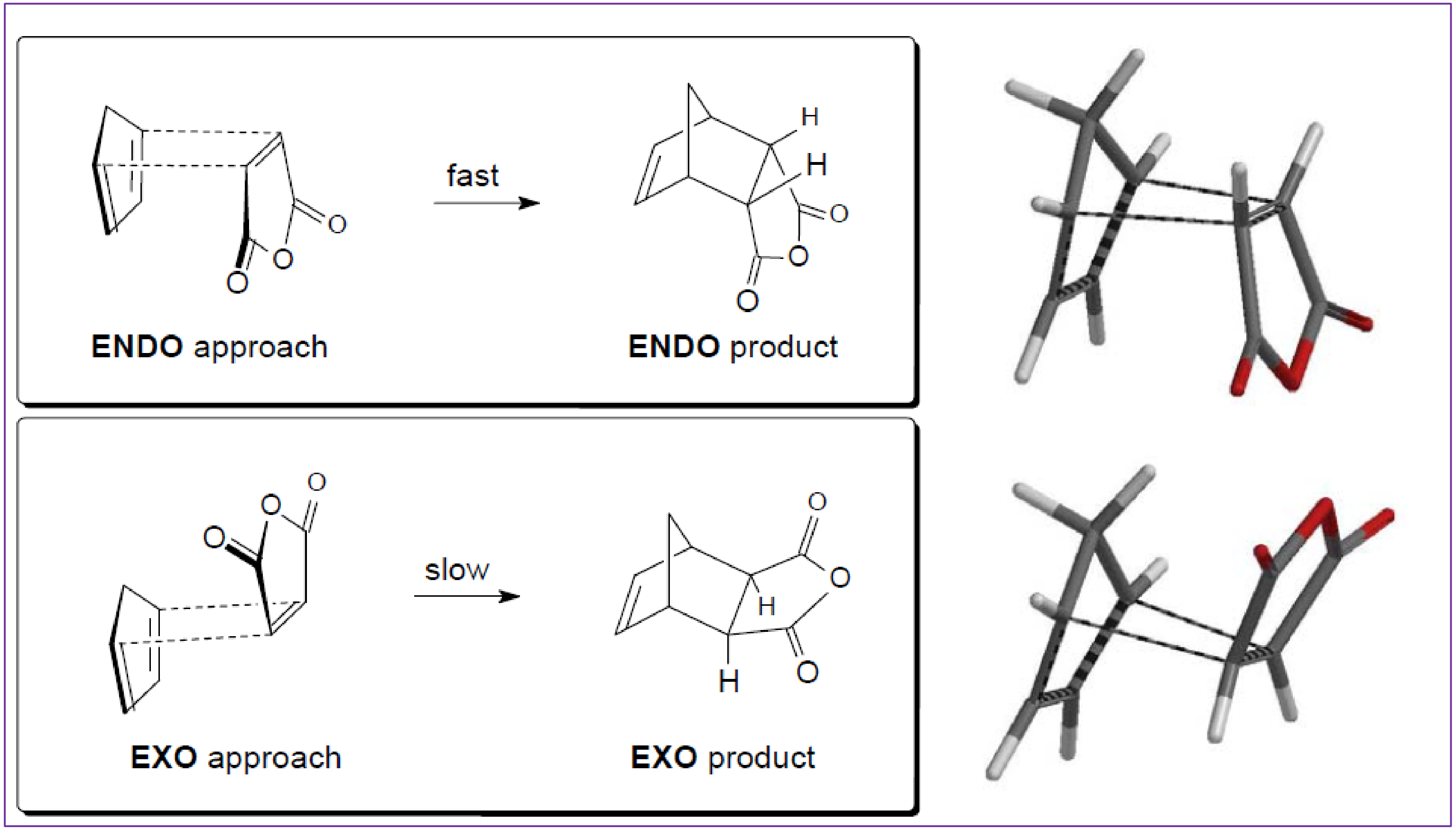
In a thermal [2 + 2] cycloaddition, like phases of the *p* orbitals on only one set of terminal carbons can overlap. For like phases to overlap on the other terminal carbon, the molecule must twist to allow for an antarafacial pathway. This process cannot occur to form small rings.



In a photochemical [2 + 2] cycloaddition, light energy promotes an electron from the ground state HOMO to form the excited state HOMO. Interaction of this excited state HOMO with the LUMO of the second alkene then allows for overlap of the like phases of both sets of p orbitals. Two bonding interactions result and the reaction occurs by a suprafacial pathway.



Diels-Alder Reaction: ENDO/EXO Products

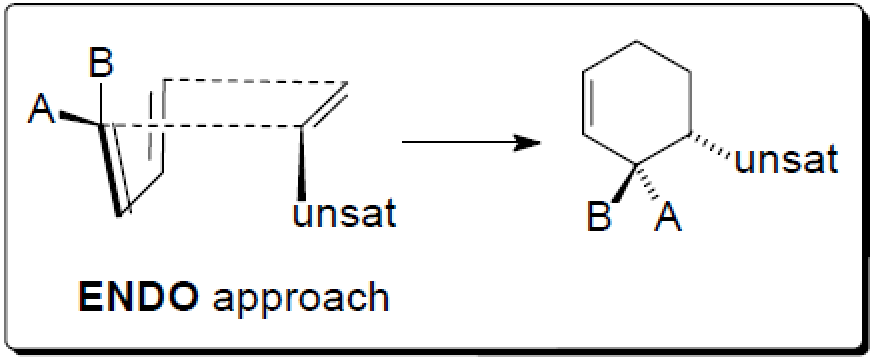


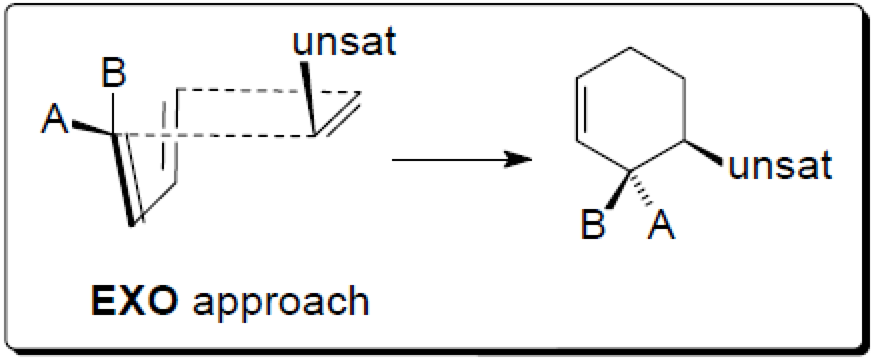
The ***endo rule*** for predicting Diels-Alder stereochemistry applies to reactions involving a dienophile with one or more unsaturated substituents. The rule states that the kinetic product is obtained from a transition state in which the dienophile substituent is “endo” with respect to the diene.

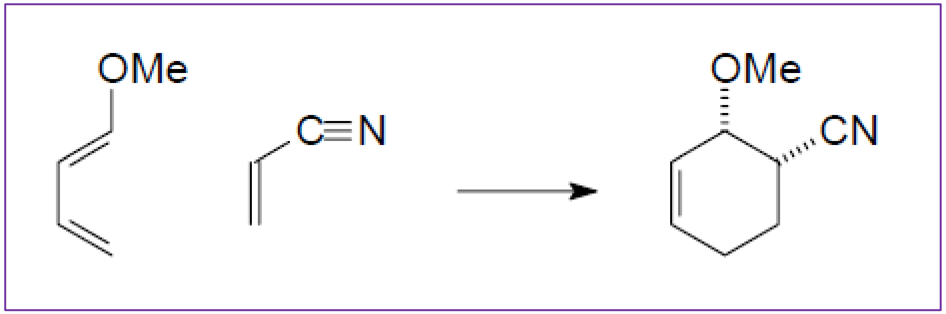
Diels-Alder Reaction: ENDO/EXO Products

In the case of a cyclic diene, like the one shown above, the meaning of "endo" is straightforward and so is application of the *endo rule*. You can just draw products with *endo* unsaturated substituents (that is, *endo* with respect to the alkene bridge).

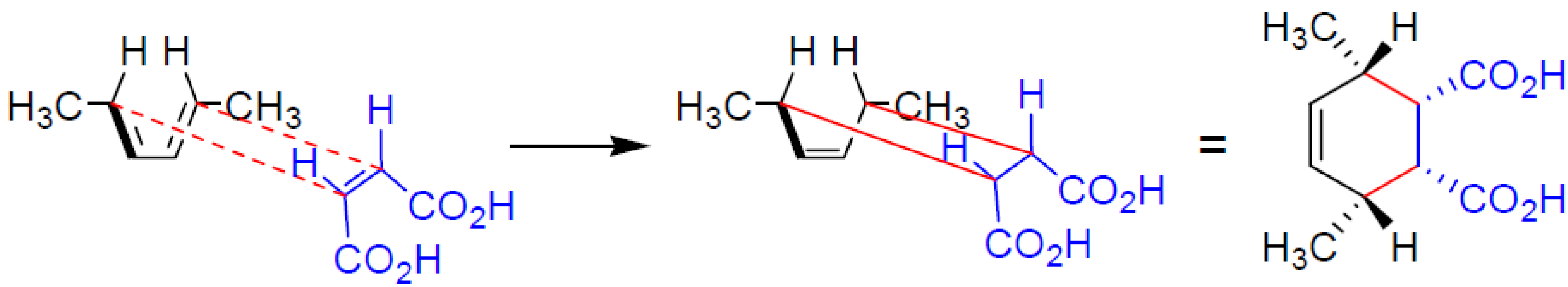
The interpretation of "endo" is far less obvious for acyclic dienes. In such cases, one needs to compare the two approaches and notice the orientation of the substituent relative to the diene. The *endo approach* places the substituent over (or *cis* to) the diene.





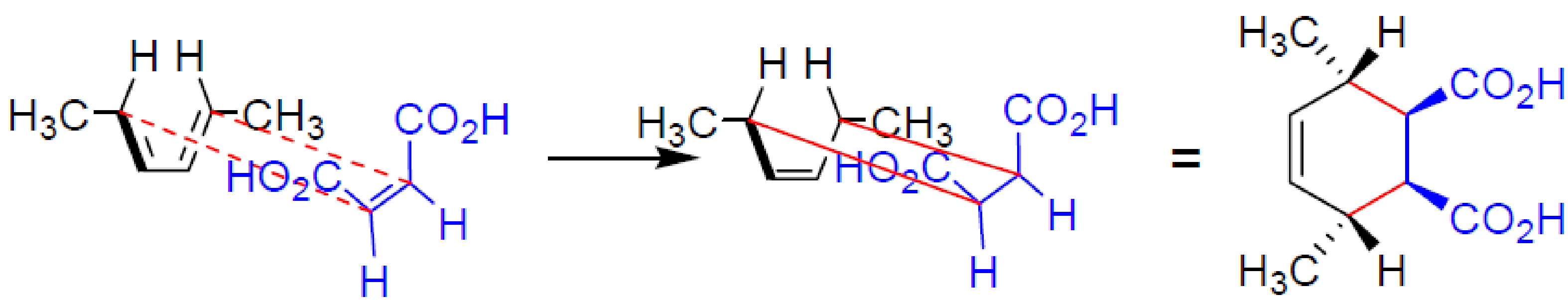


Diels-Alder Reaction: ENDO/EXO Products



ENDO transition state

ENDO product
(favored)



EXO transition state

EXO product
(disfavored)